

```

#include <stdio.h>
#include <stdlib.h>

/* Definition of node */
struct node {
    int data;
    struct node *next;
};

struct node *head = NULL;

/* Function Prototypes */
void createe();
void insertw9();
void deletee();
void traverse();

/* Main Function */
int main() {
    int choice;

    while (1) {
        printf("\n--- Singly Linked List Menu ---\n");
        printf("1. Create\n");
        printf("2. Insertw9\n");
        printf("3. Delete\n");
        printf("4. Traverse\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: createe();
                    break;
            case 2: insertw9();
                    break;
            case 3: deletee();
                    break;
            case 4: traverse();
                    break;
            case 5: exit(0);
                    default: printf("Invalid choice\n");
        }
    }
    return 0;
}

```

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/* Create Linked List */
void createe() {
    int n, i, value;
    struct node *temp, *newnode;

    printf("Enter number of nodes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        newnode = (struct node *)malloc(sizeof(struct node));
        printf("Enter data: ");
        scanf("%d", &value);

        newnode->data = value;
        newnode->next = NULL;

        if (head == NULL) {
            head = newnode;
        } else {
            temp = head;
            while (temp->next != NULL)
                temp = temp->next;
            temp->next = newnode;
        }
    }
}

```

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/* Insert at End */
void insertw9() {
    int value;
    struct node *newnode, *temp;

    newnode = (struct node *)malloc(sizeof(struct node));
    printf("Enter value to insert: ");
    scanf("%d", &value);

    newnode->data = value;
    newnode->next = NULL;

    if (head == NULL) {
        head = newnode;
    } else {
        temp = head;
        while (temp->next != NULL)
            temp = temp->next;
        temp->next = newnode;
    }
}

```

```

}

/* Delete a Node */
void deletee() {
    int value;
    struct node *temp, *prev;

    if (head == NULL) {
        printf("List is empty\n");
        return;
    }

    printf("Enter value to delete: ");
    scanf("%d", &value);

    temp = head;

    if (temp->data == value) {
        head = temp->next;
        free(temp);
        printf("Node deleted\n");
        return;
    }

    while (temp != NULL && temp->data != value) {
        prev = temp;
        temp = temp->next;
    }

    if (temp == NULL) {
        printf("Value not found\n");
    } else {
        prev->next = temp->next;
        free(temp);
        printf("Node deleted\n");
    }
}

/* Traverse the List */
void traverse() {
    struct node *temp;

    if (head == NULL) {
        printf("List is empty\n");
        return;
    }

    temp = head;

```

```
printf("Linked List: ");  
while (temp != NULL) {  
    printf("%d -> ", temp->data);  
    temp = temp->next;  
}  
printf("NULL\n");  
}
```

Compile Result

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 2

Enter value to insert: 3

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice:



GIF



1

2

3

4

5

6

7

8

9

0

q

w

e

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o

p

Compile Result

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 3

List is empty

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice:



GIF



1

2

3

4

5

6

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8

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0

q

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p

Compile Result

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 1

Enter number of nodes: 2

Enter data: 3

Enter data: 4

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 3

Enter value to delete: 4

Node deleted

--- Singly Linked List Menu ---



GIF



1

2

3

4

5

6

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p

Compile Result

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 1

Enter number of nodes: 3

Enter data: 5

Enter data: 6

Enter data: 7

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 4

Linked List: 5 -> 6 -> 7 -> NULL

--- Singly Linked List Menu ---



GIF



1

2

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Compile Result

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice: 1

Enter number of nodes: 3

Enter data: 1

Enter data: 2

Enter data: 3

--- Singly Linked List Menu ---

1. Create
2. Insertw9
3. Delete
4. Traverse
5. Exit

Enter your choice:



GIF



1

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